SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answers

Course Code:	ITA0328	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Short Answers
Weightage:	30 % of CIA	**Total Marks:**	100 Marks
CO1 Statement:	*Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
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Section / Batch:	AIML	**Date:**	

Code of Conduct

I B V Chaitanya Reddy(192425376) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Chaitanya

Instructions

1. This test contains 50 questions, Total: 100 Marks.
2. Duration: 1 hour.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Questions

Q1. A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. The planning team wants to know how many full-duplex voice channels this spectrum can support.

Q2. A regional carrier runs its network on 560 total voice channels, shared across a 7-cell frequency reuse cluster. During capacity planning, an engineer needs to know how many channels a single cell will typically be allocated.

Q3. A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels this deployment can offer.

Q4. During a site audit, engineers note the base station radiates 80 W while a nearby handset transmits just 4 W. They want to express this power gap as a decibel-based power control ratio ($10 \cdot \log 10$ of the ratio).

Q5. A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. The RF team needs the Carrier-to-Interference ratio (C/I) in dB to assess link quality.

Q6. An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7-cell pattern. How many channels does a single cell end up with?

Q7. A transmitter delivers 64 W of power to a receiver. Engineers then move the receiver twice as far away, in an environment with a path-loss exponent of 4. What received power should they now expect?

Q8. Planners want to double a cell's radius while keeping the same SNR at the new edge, in an environment obeying an inverse-square power law. By what factor should transmit power be scaled up?

Q9. A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the entire cluster?

Q10. A drive test records a vehicle traveling at 100 km/h while camped on a 900 MHz carrier. Estimate the Doppler shift the receiver experiences.

Q11. A TDMA frame lasting 5 ms is divided into 8 equal time slots. What is the duration of a single slot?

Q12. Spectrum planners have 24 MHz of allocated bandwidth and fix carrier spacing at 200 kHz for a GSM rollout. How many carrier frequencies can be created from this allocation?

Q13. A handset transmits at 8 W, but the signal experiences 33 dB of path loss on its way to the base station. Estimate the power the base station receiver actually sees, in dBm.

Q14. A network engineer uses 1 m as the reference distance while assessing a 1900 MHz deployment, where cell sites are spaced 2.5 km apart. Estimate the free-space path loss over that distance.

Q15. A cluster consists of 6 cells, each able to carry up to 250 simultaneous calls at peak. What is the cluster's total call-handling capacity?

Q16. A switching subsystem is rated to process 4000 new call requests every second, and calls last 3 minutes on average. How many calls could be active in the system at any one time?

Q17. Engineers assign 150 channels to a base station subsystem, each channel occupying 25 kHz. What total bandwidth does this base station consume?

Q18. Spectrum planners insert a 40 kHz guard band between each of 30 adjacent frequency bands to prevent interference. How much total bandwidth ends up wasted because of these guard bands?

Q19. A vehicle cruises at 90 km/h through a region of 1.5 km-radius cells. Using the cell diameter as the crossing distance, roughly how many handoffs would occur over an hour of travel?

Q20. A carrier's network hierarchy has 8 regional centers, each overseeing 40 base stations, with each base station serving 250 users. What is the network's total user-handling capacity?

Q21. Coverage planners want to blanket a 400 sq km region using hexagonal cells of 400 m radius (approximate cell area as R^2). Roughly how many cells will the rollout require?

Q22. An FDMA system allots 30 kHz to each channel, drawing from a total spectrum pool of 6 MHz. How many channels can be carved out of this allocation?

Q23. A market study notes that 1200 subscribers each place 4 calls per hour, averaging 2.5 minutes per call. What total offered traffic, in Erlangs, does this represent?

Q24. A 2.5 GHz deployment has 40 MHz of bandwidth, and each connected user consumes 400 kHz. What is the maximum number of users the band can serve at once?

Q25. A carrier's base stations each cap out at 600 simultaneous users, and the network comprises 1200 such stations. What total user capacity does the network offer?

Q26. A paging scheme sweeps through 4 cells at a time across a 200-cell network. In the worst case, how many paging rounds are needed to locate a subscriber?

Q27. A core network is engineered for 100,000 simultaneous connections. During a busy period it receives 40,000 call attempts per minute, each lasting 3 minutes on average. What fraction of the offered traffic ends up blocked?

Q28. A network policy lets 15% of its 600,000 subscribers roam onto partner networks each hour. At any given moment, how many subscribers are likely roaming?

Q29. A switching center is capable of processing 12,000 calls per second, but is currently handling a peak load of 9,000 calls per second. What percentage of its capacity is being utilized?

Q30. An operator reuses frequencies with a factor of 4 across a network holding 48 MHz of spectrum, with 200 kHz per channel. How many channels land in each cell?

Q31. A 900 MHz GSM deployment uses 200 kHz channel spacing across a 30 MHz allocation. How many channels can this spectrum support?

Q32. Over the course of an hour, a network logs 2500 handoff attempts, 99% of which succeed. How many calls were successfully handed off?

Q33. A mobile handset radiates 12 W, and the link experiences 45 dB of path loss before reaching the base station. What power level does the base station receiver detect?

Q34. A 7-cell cluster is deployed where each cell can serve up to 900 simultaneous users. What is the cluster's overall user capacity?

Q35. A TDMA carrier's frame spans 4.8 ms and is split into 8 equal slots. How long is each individual slot?

Q36. A mobile device moves at 150 km/h while connected to a 1.8 GHz carrier. What Doppler shift should the receiver expect?

Q37. A core switch is dimensioned for 250,000 simultaneous calls. It is fielding calls at a rate of 70,000 per minute, each averaging 4 minutes in duration. What share of the offered load gets blocked?

Q38. A hexagonal layout groups 9 cells into a cluster, and each cell can host 600 users. What is the cluster's combined user capacity?

Q39. A switching center processes 60,000 calls per second, with 0.4% of them dropped due to network issues. How many calls are being dropped every second?

Q40. A base station in an 1800 MHz network transmits at 43 dBm. At the edge of its 2 km-radius cell, the received signal measures -87 dBm. What path loss has the signal undergone?

Q41. A single base station covers 2.5 km² of territory, and the operator plans to deploy 150 identical stations. What total geographic area will the rollout cover?

Q42. An operator applies a frequency reuse factor of 6 over a 48 MHz spectrum allocation, with each channel taking up 200 kHz. How many channels does each cell get?

Q43. A paging system processes 6000 users per cycle, and the subscriber base totals 150,000 users. How many paging cycles are needed to sweep the entire base?

Q44. A phone call runs for 9 minutes, and during that time the network hands the call off roughly every 3 minutes. How many handoffs took place over the call?

Q45. A receiver picks up a wanted signal at -85 dBm against a noise floor of -115 dBm. What signal-to-noise ratio, in dB, does this represent?

Q46. A 4G cell is serving 250 users simultaneously, each needing a guaranteed 150 kbps. What is the minimum bandwidth the cell must provide to support them all?

Q47. A base station handles 1500 subscribers, each averaging 4 calls per hour with a 2.5-minute call length. What total traffic, in Erlangs, does this generate?

Q48. A network reserves 250 separate control channels, each 12 kHz wide, for signaling. What total bandwidth is tied up in control channels?

Q49. A network is carrying 1500 simultaneous connections, each streaming at 600 kbps on average. What total throughput must the network support?

Q50. A telecom operator wants each tower to cover a 4 km radius (treating coverage as a circle, area = πR^2) and needs to blanket a 350 km² service region. Roughly how many towers will the rollout require?

ANSWERS

Q1.

Formula: Number of simplex channels = Total spectrum ÷ Channel bandwidth; one full-duplex channel needs 2 simplex channels

Solution:

Step 1: Total spectrum = 45 MHz = 45,000 kHz

Step 2: Simplex channels = 45,000 ÷ 25 = 1,800

Step 3: Duplex channels = 1,800 ÷ 2 = 900

Answer: 900 full-duplex voice channels

Q2.

Formula: Channels per cell = Total channels ÷ Cluster size (N)

Solution:

Step 1: Channels per cell = 560 ÷ 7 = 80

Answer: 80 channels per cell

Q3.

Formula: Total channels = Number of carriers × Time slots per carrier

Solution:

Step 1: Total = 100 × 8 = 800

Answer: 800 simultaneous voice channels (the 200 kHz spacing is not needed)

Q4.

Formula: Power ratio (dB) = 10·log10(P1 / P2)

Solution:

Step 1: Ratio = 80 ÷ 4 = 20

Step 2: 10·log10(20) = 10 × 1.301 = 13.01 dB

Answer: ≈ 13 dB

Q5.

Formula: C/I (dB) = Signal level (dBm) − Interference level (dBm)

Solution:

Step 1: C/I = (−60) − (−92) = −60 + 92 = 32 dB

Answer: C/I = 32 dB

Q6.

Formula: Total channels = Spectrum ÷ Channel BW; Channels per cell = Total channels ÷ N

Solution:

Step 1: Total channels = 42 MHz ÷ 300 kHz = 42,000 ÷ 300 = 140

Step 2: Channels per cell = 140 ÷ 7 = 20

Answer: 20 channels per cell

Q7.

Formula: $Pr \propto 1 / d^n$ (n = path-loss exponent)

Solution:

Step 1: Distance doubled, so d2 = 2·d1 and n = 4

Step 2: Pr2 = Pr1 × (d1/d2)^4 = 64 × (1/2)^4 = 64 ÷ 16 = 4 W

Answer: 4 W

Q8.

Formula: Inverse-square law: $Pr \propto Pt / R^2$ (to keep the same received level, $Pt \propto R^2$)

Solution:

Step 1: R2 = 2·R1

Step 2: $Pt2 / Pt1 = (R2 / R1)^2 = 2^2 = 4$

Answer: Transmit power must be increased by a factor of 4

Q9.

Formula: Cluster channels = Cluster size × Channels per cell

Solution:

Step 1: Total = 4 × 15 = 60

Answer: 60 channels

Q10.

Formula: Doppler shift $f_d = v / \lambda = v \cdot f / c$

Solution:

Step 1: $v = 100$ km/h $= 100 \times 1000 / 3600 = 27.78$ m/s

Step 2: $\lambda = c / f = (3 \times 10^8) / (900 \times 10^6) = 0.333$ m

Step 3: $f_d = 27.78 \div 0.333 = 83.3$ Hz

Answer: ≈ 83 Hz (maximum Doppler shift)

Q11.

Formula: Slot duration = Frame duration ÷ Number of slots

Solution:

Step 1: Slot = 5 ms ÷ 8 = 0.625 ms

Answer: 0.625 ms (625 µs)

Q12.

Formula: Number of carriers = Total bandwidth ÷ Carrier spacing

Solution:

Step 1: 24 MHz = 24,000 kHz

Step 2: Carriers = 24,000 ÷ 200 = 120

Answer: 120 carrier frequencies

Q13.

Formula: P_r (dBm) = P_t (dBm) − Path loss (dB)

Solution:

Step 1: $P_t = 8$ W $= 8000$ mW $\rightarrow 10 \cdot \log10(8000) = 39.03$ dBm

Step 2: $P_r = 39.03 - 33 = 6.03$ dBm

Answer: ≈ 6 dBm (about 4 mW)

Q14.

Formula: Free-space path loss: $PL(d) = PL(d0) + 20 \cdot \log10(d / d0)$, where $PL(d0) = 20 \cdot \log10(4\pi \cdot d0 / \lambda)$

Solution:

Step 1: $\lambda = c / f = (3 \times 10^8) / (1.9 \times 10^9) = 0.1579$ m

Step 2: $PL(d0 = 1 \text{ m}) = 20 \cdot \log10(4\pi \times 1 / 0.1579) = 20 \cdot \log10(79.6) = 38.0$ dB

Step 3: $20 \cdot \log10(d / d0) = 20 \cdot \log10(2500 / 1) = 67.96$ dB

Step 4: $PL = 38.0 + 67.96 = 105.98$ dB

Answer: ≈ 106 dB

Q15.

Formula: Cluster capacity = Cells in cluster × Calls per cell

Solution:

Step 1: Capacity = 6 × 250 = 1,500

Answer: 1,500 simultaneous calls

Q16.

Formula: Little's Law: Active calls = Arrival rate × Average holding time

Solution:

Step 1: Holding time = 3 min = 180 s

Step 2: Active calls = 4000 × 180 = 720,000

Answer: 720,000 simultaneous calls

Q17.

Formula: Total bandwidth = Number of channels × Bandwidth per channel

Solution:

Step 1: 150 × 25 kHz = 3,750 kHz = 3.75 MHz

Answer: 3.75 MHz

Q18.

Formula: Wasted bandwidth = Number of guard bands × Guard-band width

Solution:

Step 1: 30 × 40 kHz = 1,200 kHz = 1.2 MHz

Answer: 1.2 MHz (note: if only the 29 gaps strictly between 30 bands are counted, it is 29 × 40 = 1.16 MHz)

Q19.

Formula: Handoffs = Distance travelled ÷ Cell diameter

Solution:

Step 1: Distance in 1 hour = 90 km

Step 2: Cell diameter = 2 × 1.5 = 3 km

Step 3: Handoffs = 90 ÷ 3 = 30

Answer: ≈ 30 handoffs per hour

Q20.

Formula: Total users = Regions × Base stations per region × Users per base station

Solution:

Step 1: 8 × 40 × 250 = 320 × 250 = 80,000

Answer: 80,000 users

Q21.

Formula: Number of cells = Total area ÷ Area per cell

Solution:

Step 1: R = 400 m = 0.4 km

Step 2: Cell area ≈ R^2 = 0.16 km^2

Step 3: Cells = 400 ÷ 0.16 = 2,500

Answer: ≈ 2,500 cells

Q22.

Formula: Channels = Total spectrum ÷ Channel bandwidth

Solution:

Step 1: 6 MHz = 6,000 kHz

Step 2: Channels = 6,000 ÷ 30 = 200

Answer: 200 channels

Q23.

Formula: Traffic (Erlangs) = (Number of calls per hour × Average call duration in min) ÷ 60

Solution:

Step 1: Calls per hour = 1200 × 4 = 4,800

Step 2: Total minutes = 4,800 × 2.5 = 12,000 min

Step 3: Erlangs = 12,000 ÷ 60 = 200

Answer: 200 Erlangs

Q24.

Formula: Maximum users = Total bandwidth ÷ Bandwidth per user

Solution:

Step 1: 40 MHz = 40,000 kHz

Step 2: Users = 40,000 ÷ 400 = 100

Answer: 100 users (the 2.5 GHz carrier frequency is not needed)

Q25.

Formula: Network capacity = Number of base stations × Users per base station

Solution:

Step 1: 1200 × 600 = 720,000

Answer: 720,000 users

Q26.

Formula: Paging rounds (worst case) = Total cells ÷ Cells paged per round

Solution:

Step 1: 200 ÷ 4 = 50

Answer: 50 paging rounds

Q27.

Formula: Offered load (Erlangs) = Call arrival rate × Holding time; Blocking ≈ (Offered − Capacity) ÷

Offered

Solution:

Step 1: Offered load = 40,000 calls/min × 3 min = 120,000 Erlangs

Step 2: Capacity = 100,000 connections

Step 3: Excess load = 120,000 − 100,000 = 20,000

Step 4: Blocked fraction = 20,000 ÷ 120,000 = 0.1667

Answer: ≈ 16.7 % of the offered traffic is blocked

Q28.

Formula: Roaming subscribers = Percentage × Total subscribers

Solution:

Step 1: 0.15 × 600,000 = 90,000

Answer: 90,000 subscribers

Q29.

Formula: Utilization (%) = (Current load ÷ Capacity) × 100

Solution:

Step 1: (9,000 ÷ 12,000) × 100 = 75 %

Answer: 75 % utilized

Q30.

Formula: Channels per cell = (Total spectrum ÷ Channel BW) ÷ Reuse factor N

Solution:

Step 1: Total channels = 48 MHz ÷ 200 kHz = 48,000 ÷ 200 = 240

Step 2: Channels per cell = 240 ÷ 4 = 60

Answer: 60 channels per cell

Q31.

Formula: Channels = Total spectrum ÷ Channel spacing

Solution:

Step 1: 30 MHz = 30,000 kHz

Step 2: Channels = 30,000 ÷ 200 = 150

Answer: 150 channels

Q32.

Formula: Successful handoffs = Handoff attempts × Success rate

Solution:

Step 1: 2500 × 0.99 = 2,475

Answer: 2,475 successful handoffs

Q33.

Formula: Pr (dBm) = Pt (dBm) − Path loss (dB)

Solution:

Step 1: Pt = 12 W = 12,000 mW → 10·log10(12,000) = 40.79 dBm

Step 2: Pr = 40.79 − 45 = −4.21 dBm

Step 3: In watts: Pr = 12 ÷ 10^4.5 ≈ 3.8×10^{-4} W (0.38 mW)

Answer: ≈ −4.2 dBm (≈ 0.38 mW)

Q34.

Formula: Cluster capacity = Cells × Users per cell

Solution:

Step 1: 7 × 900 = 6,300

Answer: 6,300 users

Q35.

Formula: Slot duration = Frame duration ÷ Number of slots

Solution:

Step 1: 4.8 ms ÷ 8 = 0.6 ms

Answer: 0.6 ms (600 µs)

Q36.

Formula: Doppler shift fd = v / λ

Solution:

Step 1: v = 150 km/h = 150 × 1000 / 3600 = 41.67 m/s

Step 2: $\lambda = (3\times10^{8}) / (1.8\times10^{9}) = 0.1667$ m

Step 3: fd = 41.67 ÷ 0.1667 = 250 Hz

Answer: 250 Hz

Q37.

Formula: Offered load (Erlangs) = Call rate × Holding time; Blocking ≈ (Offered − Capacity) ÷ Offered

Solution:

Step 1: Offered load = 70,000 calls/min × 4 min = 280,000 Erlangs

Step 2: Capacity = 250,000 calls

Step 3: Excess = 280,000 − 250,000 = 30,000

Step 4: Blocked share = 30,000 ÷ 280,000 = 0.1071

Answer: ≈ 10.7 % of the offered load is blocked

Q38.

Formula: Cluster capacity = Cells in cluster × Users per cell

Solution:

Step 1: 9 × 600 = 5,400

Answer: 5,400 users

Q39.

Formula: Dropped calls per second = Call rate × Drop percentage

Solution:

Step 1: 60,000 × 0.4 % = 60,000 × 0.004 = 240

Answer: 240 calls dropped per second

Q40.

Formula: Path loss (dB) = Transmit power (dBm) − Received power (dBm)

Solution:

Step 1: PL = 43 − (−87) = 43 + 87 = 130 dB

Answer: 130 dB

Q41.

Formula: Total area = Number of stations × Area per station

Solution:

Step 1: 150 × 2.5 km² = 375 km²

Answer: 375 km²

Q42.

Formula: Channels per cell = (Total spectrum ÷ Channel BW) ÷ Reuse factor N

Solution:

Step 1: Total channels = 48 MHz ÷ 200 kHz = 240

Step 2: Channels per cell = 240 ÷ 6 = 40

Answer: 40 channels per cell

Q43.

Formula: Paging cycles = Total subscribers ÷ Users per cycle

Solution:

Step 1: 150,000 ÷ 6,000 = 25

Answer: 25 paging cycles

Q44.

Formula: Number of handoffs = Call duration ÷ Handoff interval

Solution:

Step 1: 9 min ÷ 3 min = 3

Answer: 3 handoffs (roughly)

Q45.

Formula: SNR (dB) = Signal power (dBm) − Noise power (dBm)

Solution:

Step 1: SNR = (−85) − (−115) = 30 dB

Answer: SNR = 30 dB

Q46.

Formula: Minimum bandwidth (capacity) = Number of users × Rate per user

Solution:

Step 1: 250 × 150 kbps = 37,500 kbps = 37.5 Mbps

Answer: 37.5 Mbps

Q47.

Formula: Traffic (Erlangs) = (Number of calls per hour × Average call duration in min) ÷ 60

Solution:

Step 1: Calls per hour = 1500 × 4 = 6,000

Step 2: Total minutes = 6,000 × 2.5 = 15,000 min

Step 3: Erlangs = 15,000 ÷ 60 = 250

Answer: 250 Erlangs

Q48.

Formula: Total bandwidth = Number of channels × Bandwidth per channel

Solution:

Step 1: 250 × 12 kHz = 3,000 kHz = 3 MHz

Answer: 3 MHz

Q49.

Formula: Total throughput = Number of connections × Rate per connection

Solution:

Step 1: 1500 × 600 kbps = 900,000 kbps = 900 Mbps

Answer: 900 Mbps

Q50.

Formula: Number of towers = Service area ÷ Area per tower, with area per tower = $\pi \cdot R^2$

Solution:

Step 1: Area per tower = $\pi \times 4^2 = 50.27$ km²

Step 2: Towers = 350 ÷ 50.27 = 6.96

Step 3: Round up to cover the whole region

Answer: ≈ 7 towers